

Multimedia systems; wireless and mobile communication

One college period · 40-45 minutes · Multimedia and wireless

What you will learn

- Define multimedia components and file types
- Use layers, keyframes, export/share a short video (theory + lab idea)
- Compare Wi-Fi, Bluetooth, infrared, WiMAX, microwave
- Outline cellular network, 2G-5G, GEO/MEO/LEO, ad hoc / MANET

Multimedia. The combination of text, graphics, animation, audio and video used to communicate. Advantages: richer teaching, advertising, games, training.

Component	Typical files
Text	.txt .doc .pdf
Graphics	.jpg .png .gif .bmp .svg
Animation	.gif .mp4
Audio	.mp3 .wav .aac
Video	.mp4 .avi .mkv

Packages: web apps vs desktop (ProShow Gold, JetAudio, Shotcut, Canva, PowerPoint). Scene/canvas = work area; layers stack pictures/text; keyframes mark motion in animation. Lab: 2-minute video, export MP4, upload/share.

Wireless

- Wireless: no physical cable; uses electromagnetic waves. Mobile: communication while moving.
- Terms: frequency spectrum, radio signal, transceiver, access point, line of sight.

Short range	Typical use
Wi-Fi	LAN internet in college / home
Bluetooth	earbuds, file share, few metres
Infrared	TV remote, line of sight
Microwave	dishes, backhaul
WiMAX	city-wide wireless (MAN-like)

Satellites: GEO ~36000 km (TV), MEO (GPS), LEO (phones, Starlink-style). Cellular: cell, BTS, BSC, MSC, frequency reuse. Generations: 2G GSM voice, 3G GPRS/data, 4G LTE fast data, 5G low latency. Ad hoc: no fixed AP; MANET is a mobile ad hoc network.

اردو میں سمجھیں

ملٹی میڈیا متن، تصویر، آواز، ویڈیو، اینیمیشن کا ملاپ ہے۔ وائرلیس تار کے بغیر ریڈیو لہریں: وائی فائی، بلوٹوتھ، سیلولر 2G سے 5G۔

CHECK YOURSELF

1. Bluetooth vs Wi-Fi range?
2. 4G technology name often asked?

Answers: 1. Bluetooth shorter; Wi-Fi room/building · 2. LTE

MCQs

1. MP3 is mainly a file of:
 - a. video
 - b. audio
 - c. database
 - d. source code
2. Wi-Fi is a:
 - a. wired LAN cable
 - b. short-range wireless technology
 - c. printer type
 - d. CPU register
3. GSM is associated with:
 - a. 2G
 - b. 5G only
 - c. fibre optics
 - d. OSI layer 7 only

Key: 1=b, 2=b, 3=a

Short questions (3 marks each)

1. Define multimedia and its components.
2. Differentiate Wi-Fi and Bluetooth.
3. What is a cellular network? Name 2G to 5G.
4. GEO vs LEO satellites.

Model short answers

1. Combination of text, graphics, animation, audio, video. Used in education and ads.
2. Wi-Fi: higher speed/range for internet. Bluetooth: very short range for accessories.
3. Geographic area split into cells with BTS. 2G GSM, 3G, 4G LTE, 5G.
4. GEO high and stationary over earth (TV). LEO low and fast (phones, imaging).

Long questions (board Section C)

1. Explain multimedia components and how you would make a 2-minute educational video. OR Explain wireless technologies and cellular generations.